Functional Low Poly Toilet

Animated Flush + Water

Version 1.0 EZPZ Games

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1. Overview

A stylized low poly toilet with a fully functional flush animation. The flush includes an animated handle, draining and refilling water driven by blend shapes, and synced audio.

The asset is ready to use out of the box. Drop the prefab into your scene and call a single method to trigger the flush.

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2. Requirements

- Unity 2021.3 or newer
- Universal Render Pipeline (URP)

This package has no third-party dependencies. It does not require TextMeshPro, the Input System package, or any external asset.

Note: materials use URP shaders. The asset will render incorrectly in the Built-in or High Definition render pipelines without manual material conversion.

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3. Quick Start

Step 1. Import the package into your URP project.

Step 2. Locate the prefab at:

`Assets/EZPZ Games/FunctionalBathroom/Toilet/Prefabs/Toilet.prefab`

Step 3. Drag the prefab into your scene.

Step 4. Trigger the flush from your own code:

```
using EZPZGames.FunctionalBathroom;

public class MyInteraction : MonoBehaviour
{
    [SerializeField] private ToiletController toilet;

    public void OnPlayerInteract()
    {
        toilet.Flush();
    }
}
```

That is the entire integration. The controller handles the animation, the audio, and the state guarding internally.

Alternative: wire a UI Button's OnClick event directly to the `ToiletController.Flush()` method in the Inspector, with no code at all.

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4. Package Contents

```
EZPZ Games/
└── FunctionalBathroom/
    └── Toilet/
        ├── Animations/    Flush animation clip and animator
controller
        ├── Audio/         Flush sound effect
        ├── Demo/          Example scene
        ├── Materials/     URP materials
        ├── Models/        Toilet FBX
        ├── Prefabs/       Ready-to-use prefab
        └── Scripts/       ToiletController.cs
```

Demo scene: `Toilet/Demo/Toilet_Demo.unity`

Open the scene, enter Play Mode, and click the on-screen button to flush.

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5. Component Reference

ToiletController

Attached to the root of the toilet prefab. Requires an Animator on the same GameObject.

Inspector fields

Field	Description
Flush State Name	Name of the flush state in the Animator Controller. Default: `Flush`
Flush Speed	Playback speed multiplier, 0.5 to 2.0. Default: 1.0
Audio Source	The AudioSource used for flush audio
Flush Clip	The audio clip played on flush
Sync Audio Pitch	When enabled, audio pitch scales with Flush Speed so sound stays in time with the animation

Public methods

Method	Description
`Flush()`	Starts the flush. Ignored if a flush is already in progress.
`SetFlushSpeed(float speed)`	Changes playback speed. Clamped between 0.5 and 2.0.
`CancelFlush()`	Stops the flush and its audio immediately.

Public properties

Property	Description
`IsFlushing`	Returns true while a flush is running.
`FlushSpeed`	Returns the current speed

Property	Description
	multiplier.

Events

Event	Fires when
`OnFlushStarted`	The flush begins
`OnFlushCompleted`	The flush animation finishes

Both are UnityEvents and can be wired in the Inspector or subscribed to in code.

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6. Common Tasks

Changing the flush duration

The flush runs for approximately 14 seconds at default speed. To change it:

```
toilet.SetFlushSpeed(1.5f);   // 50% faster
toilet.SetFlushSpeed(0.75f);  // 25% slower
```

Leave **Sync Audio Pitch** enabled to keep the sound aligned. Speeds beyond roughly 1.3x will noticeably alter the pitch of the audio.

Reacting to the flush finishing

```
private void OnEnable()
{
    toilet.OnFlushCompleted.AddListener(HandleFlushDone);
}

private void OnDisable()
{
    toilet.OnFlushCompleted.RemoveListener(HandleFlushDone);
}

private void HandleFlushDone()
{
    // your logic here
}
```

Replacing the flush sound

Assign your own AudioClip to the **Flush Clip** field on the ToiletController. For best results, use a clip close to 14 seconds so it matches the animation length, or adjust Flush Speed to suit your clip.

Using your own water material

The water surface is a separate mesh named `Water` inside the prefab. Replace the material on its Skinned Mesh Renderer with your own. Keep the shader's Surface Type set to Transparent.

Removing the water

Disable or delete the `Water` child object. The flush animation and audio will still play. This saves approximately 450 triangles.

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7. Model Information

Item	Value
Total triangles	~1,390
Water surface	~450 triangles
Materials	3
Textures	None (solid colour materials)
Animated parts	Flush handle, water surface

Prefab hierarchy

```
Toilet                  Root, holds ToiletController and Animator
├── ToiletBase
├── TankCover
├── FlushHandle         Animated
├── ToiletSeat
├── ToiletSeatCover
├── BackPipe
└── Water               Blend shapes: Drain, Vortex
```

All pivots are placed for practical use: the base pivot sits at floor level for easy placement, and hinged parts pivot at their hinge points.

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8. Troubleshooting

The toilet appears magenta. The project is not using the Universal Render Pipeline. If using Built-in, go to Materials folder and convert each material to Standard. If using HDRP, convert each material to HDRP/Lit. Make sure to set M_Water to Transparent under "Surface Type".

Nothing happens when I call Flush(). Check that the Animator component has the included `AC_Toilet` controller assigned, and that the **Flush State Name** field matches the state name in that controller.

The flush plays but there is no sound. Confirm an AudioSource is assigned to the controller's **Audio Source** field and that a clip is assigned to **Flush Clip**. Also check that **Play On Awake** is disabled on the AudioSource.

OnFlushCompleted never fires. This event is raised by an Animation Event on the final frame of the flush clip. If you replaced or re-authored the animation, re-add an Animation Event calling `OnFlushAnimationComplete` on the last frame.

The water looks solid rather than transparent. Select the water material and confirm Surface Type is set to Transparent and the Base Map colour has an alpha value below 255.

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9. Support

For questions, bug reports, or feature requests, contact EZPZ Games through the publisher page on the Unity Asset Store.

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